Package ‘dsHarmonization’

July 6, 2026

Title What the Package Does (One Line, Title Case)

Version 0.0.0.9000

Description What the package does (one paragraph).

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Encoding UTF-8

Roxygen list(markdown = TRUE)

Config/roxygen2/version 8.0.0

Contents

`cast_NADS` *Cast "NA" string into type NA*

Description

Convert "NA" strings of a data frame to proper NA values and report whether any replacement occurred.

Usage

```
cast_NADS(df)
```

Arguments

`df`	A data frame containing values to be converted.

Value

A list containing:

- data: the input data frame with "NA" strings converted to NA
- flag: logical indicating whether any replacement was performed

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`cast_to_factorDS` *Cast specified columns to factor*

Description

Convert one or more columns of a data frame to factor type.

Usage

```
cast_to_factorDS(df, columns)
```

Arguments

`df`	A data frame containing the columns to be converted.
`columns`	A single character string specifying the column names separated by "$".

Value

The input data frame with the specified columns converted to factors.

`check_categoricalDS` *Check validity of categorical variables*

Description

Verify that selected categorical variables in a data frame contain only predefined valid values.

Usage

```
check_categoricalDS(df)
```

Arguments

`df`	A data frame containing the categorical variables to be checked.

Value

A character vector with the names of columns containing invalid values. Returns an empty character vector if all checked variables are valid.

`check_missing_dataDS` *Compute percentage of missing values per column*

Description

Computes the proportion of missing values (NA) for each column of a data frame. The function returns a named numeric vector containing the percentage of missing values for every variable.

Usage

```
check_missing_dataDS(df)
```

Arguments

`df`	A data frame for which the percentage of missing values per column is to be calculated.

Value

A named numeric vector where each element represents the percentage of missing values in the corresponding column of `df`.

`check_numericDS` *Check validity of numeric variables*

Description

Verify that selected numeric variables in a data frame fall within predefined ranges and satisfy integer constraints where required. Also checks the format of the patient identifier (pat_ID), if present.

Usage

```
check_numericDS(df)
```

Arguments

`df`	A data frame containing the numeric variables to be checked.

Value

A character vector with the names of columns containing invalid values. Returns an empty character vector if all checked variables are valid.

check_patient_visitDS *Count patient-visit combinations*

Description

Create and count all patient-visit combinations in a data frame, grouping by patient identifier and visit time from diagnosis.

Usage

```
check_patient_visitDS(df)
```

Arguments

df	A data frame containing at least the columns pat_ID and Visit_months_from_diagnosis.

Value

A data frame with one row per patient-visit combination and a column n indicating the number of occurrences.

check_variablesDS *Check that required variables are present in a data frame*

Description

Verify that a data frame contains the expected variables.

Usage

```
check_variablesDS(df, variables_string)
```

Arguments

df	A data frame to be checked.
variables_string	A single character string specifying the expected variable names separated by "$".

Value

A named list with two elements:

- missing: variables specified but not found in the data frame.
- extra: variables found in the data frame but not specified.

`fill_missing_dataDS` *Fill missing values by patient and visit order*

Description

For each patient, sort visits by time from diagnosis and propagate non-missing values of a specified column. Missing values are replaced with the most recent non-missing value observed in chronological order within the same patient.

Usage

```
fill_missing_dataDS(df, pat_id_col, visit_col, value_col)
```

Arguments

`df`	A data frame containing at least the patient identifier column, the visit time column, and the variable to be filled.
`pat_id_col`	A character string specifying the name of the patient identifier column.
`visit_col`	A character string specifying the name of the visit time column.
`value_col`	A character string specifying the name of the column whose missing values should be filled.

Value

A data frame where missing values in `value_col` are filled forward within each patient group according to increasing visit time.

`get_cast_NA_changedDS` *Check if NA string replacement occurred*

Description

Returns whether the cast_NA operation modified the dataset.

Usage

```
get_cast_NA_changedDS(x)
```

Arguments

`x`	A DataSHIELD object returned by cast_NADS.

Value

Logical value indicating if any "NA" strings were replaced.

global_imputationDS *Global imputation of missing data using MICE*

Description

Perform a global imputation of missing values in a data frame using the Multiple Imputation by Chained Equations (MICE) algorithm. The function automatically identifies continuous and categorical variables, assigns appropriate imputation methods (pmm, logreg, polyreg), and returns a single imputed data frame.

Usage

```
global_imputationDS(df, id_col, m, maxit)
```

Arguments

`df`	A data frame containing the variables to be imputed. Both numeric and categorical variables are supported.
`id_col`	A string specifying the name of the identifier column. This column is excluded from the imputation process and used to aggregate the results across multiple imputations.
`m`	Number of imputations to generate (passed to `mice`).
`maxit`	Maximum number of MICE iterations per imputation chain.

Value

A data frame where missing values have been imputed using the MICE

remove_columnsDS *Remove specified columns from a data frame*

Description

Remove one or more columns from a data frame based on their names.

Usage

```
remove_columnsDS(df, col_names)
```

Arguments

`df`	A data frame from which columns will be removed.
`col_names`	A character vector specifying the names of the columns to be removed.

Value

The input data frame with the specified columns removed.

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